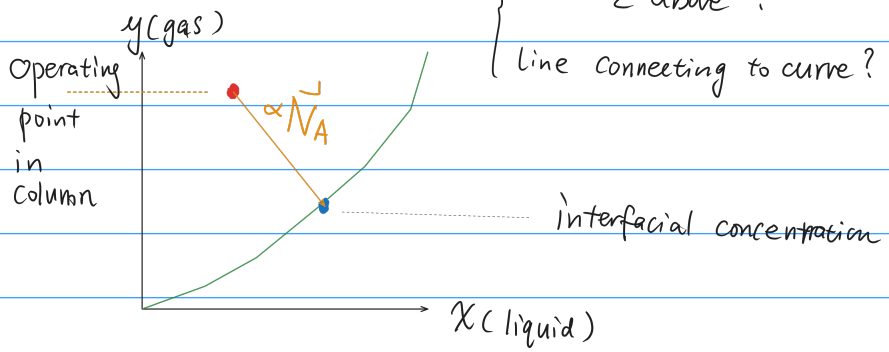
CHE318 L24

Design of packed bed absorption

Recap of last week:

① Eq diagram ? Meaning of points { on curve? above ? } line connecting to curve ?



1) For diluted (EMCD-like) system, we have

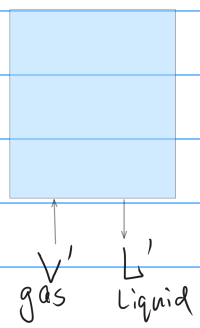
$$N_A \doteq k_y' (y_{AG} - y_{Ai}) = k_x' (x_{Ai} - x_{AL})$$

Concentrated solute through stagnant films ⇒ need to use $\frac{k_x'}{x_{Bm}}$ talked in this class

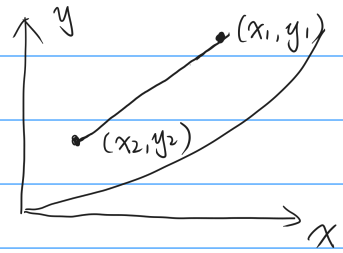
2) The series of (x, y) in tower / column forms the operating line (OL)

M.B. For operating line

$$L'\left(\frac{x_2}{1-x_2}\right) + V'\left(\frac{y_1}{1-y_1}\right) = L'\left(\frac{x_1}{1-x_1}\right) + V'\left(\frac{y_2}{1-y_2}\right)$$



What can it solve ?



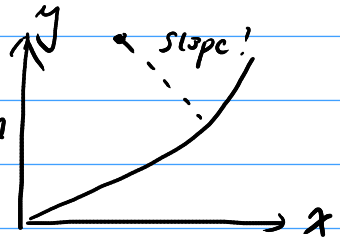
Dilute absorption tower ⇒ $y = f(x)$ is a straight line

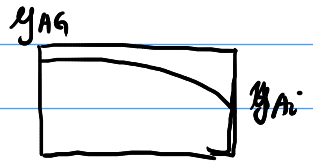
$$y = \left(\frac{L'}{V'}\right) x + [\text{const}]$$

But not linear if $1 - x \neq 1$ (see assignment 6)

More accurate description for the slope

Usually, $1-x_{AL}$ & $1-y_{AG}$ is not close enough to 1 ?



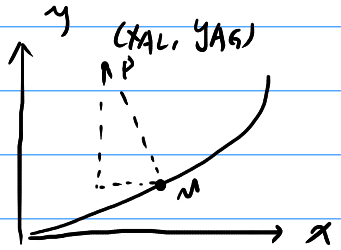


y_{AG} y_{Ai} relation is non-linear, use $x_{Bm} = (1-y)_{im}$ correction!

(Usual description: diffusion of A through non-diffusing B)

Still, mass balance of N_A across Liq ⋮ Gas is needed

$$N_A = \frac{k_y'}{(1-y)_{im}}(y_{AG} - y_{Ai}) = \frac{k_x'}{(1-x)_{im}}(x_{Ai} - x_{AL})$$



$$\text{Slope} = \frac{y_{AG} - y_{Ai}}{x_{AL} - x_{Ai}} = -\frac{k_x'/(1-x_A)_{im}}{k_y'/(1-y_A)_{im}}$$

$(1-x_A)_{im}$ & $(1-y_A)_{im}$ depends on the actual location ⇒ ① trial & error ② numerical

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Test case: A diffusing through non-stagnant B

$P = (0.100, 0.380)$

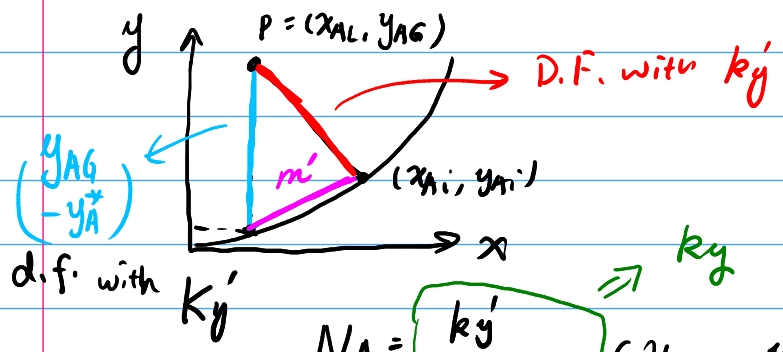
find interfacial conc? $\begin{cases} k_x' = 1.967 \times 10^{-3}\ \text{kgmol}/m^2 \cdot s \\ k_y' = 1.465 \times 10^{-3}\ \text{kgmol}/(m^2 \cdot s) \end{cases}$

Manual: $P \rightarrow$ slope $-\frac{k_x'/1}{k_y'/1}$ (x_{i1}, y_{i1})

N → update $(1-x)_{im}$, $(1-y)_{im}$

Same slope? Y → Obtained (x_{Ai}, y_{Ai})

Question 2: overall m.t. coefficient



$$N_A = \boxed{\frac{k'_y}{(1-y)_{im}}}\,(y_{AG} - y_{Ai}) = \boxed{\frac{K'_y}{(1-y)_{*m}}}\,(y_{AG} - y^*_A)$$

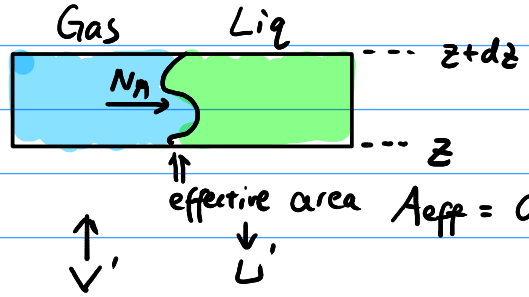
$\Rightarrow k_y$ $\Rightarrow K_y$

Relation? Geometric gives

$$\frac{1}{K_y} = \frac{1}{k_y} + \frac{m'}{k_x}$$

In-depth analysis for packed absorption tower!

Example: { Analysis of concentration profile $x(z)$, $y(z)$
Design of packed bed height z



$A_{eff} = a \cdot S \cdot dz$

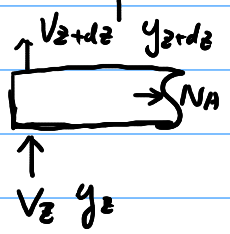
We need to solve mass balance in 1-phase!
For such control volume, we get

$$In_{liq} + In_{gas} = Out_{liq} + Out_{gas}$$

$$\underset{\frac{kgmol}{s}}{d(L\cdot x)} \Leftarrow \frac{In_{liq} - Out_{liq}}{dz} = \frac{Out_{gas} - In_{gas}}{dz} \Rightarrow d(V\cdot y)$$

$\Rightarrow$ ① Gov. Eq: $\underbrace{d(V\cdot y)}_{gas} = \underbrace{d(L\cdot x)}_{Liq}$

② Each phase mass balance gives differential eq. for $d(V\cdot y)$ & $d(L\cdot x)$

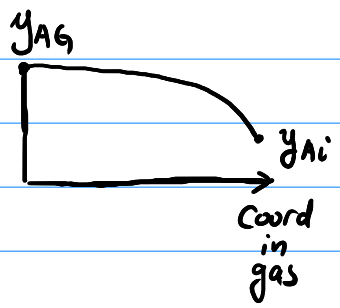


$$d(V\cdot y_{AG}) = d(A_{eff} N_A)$$

y_{AG} in bulk, to distinguish interface

$d(A_{eff} N_A)$?

$$= \frac{k_y'}{y_{Bm}} (y_{AG} - y_{Ai}) \cdot a \cdot S \cdot dz$$



y_{Bm}: log mean of $(1-y_A)$ from bulk y_{AG} → y_{Ai}

other form $(1-y_A)_{im}$ ← log mean
↑ interface

* The log-mean form is assumed for the exponential y_A profile at boundary (diffusion through non-diffusing B)

$$d(V y_{AG}) = d\left(V' \cdot \frac{y_{AG}}{1-y_{AG}}\right) = \underset{\substack{\Uparrow \\ \text{constant!}}}{V'}\, d\left(\frac{y_{AG}}{1-y_{AG}}\right)$$

Combine with expression for $d(N_A \cdot A_{eff})$, we get:

$$\frac{d}{du}\left(\frac{u}{1-u}\right) = \frac{1\cdot(1-u) - u\cdot(-1)}{(1-u)^2} = \frac{1}{(1-u)^2}$$

$$V' \underbrace{d\left(\frac{y_{AG}}{1-y_{AG}}\right)} = \frac{k_y' a}{(1-y_A)_{im}} (y_{AG} - y_{Ai})\, S\, dz$$

$$V' \frac{1}{(1-y_{AG})^2}\, \underset{\substack{\Uparrow \\ \text{variable is } y_{AG}}}{d\, y_{AG}} = \frac{k_y' a S}{(1-y_A)_{im}} (y_{AG} - \underset{\substack{\Uparrow \\ \text{where to get } y_{Ai}\text{?}}}{y_{Ai}})\, dz$$

from $\begin{cases} \text{eq curve!} \\ k_x'/k_y' \end{cases}$

Goal is to have tower height solved, so expression of Z

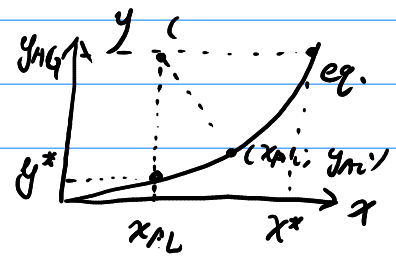
$$dz = \frac{V'}{\frac{k_y' a S}{(1-y_A)_{im}}} \cdot \frac{1}{(1-y_{AG})^2 (y_{AG} - y_{Ai})}\, d y_{AG}$$

Height $$Z = \int_0^z dz = \int_{y_2}^{y_1} \frac{V'}{\frac{k_y' a S}{(1-y_A)_{im}}} \cdot \frac{1}{(1-y_{AG})^2 (y_{AG} - y_{Ai})}\, d y_{AG}$$

Same equation for Liquid

$$Z = \int_0^z dz = \int_{x_2}^{x_1} \frac{L'}{\frac{k_x' a S}{(1-x_A)_{im}}} \cdot \frac{1}{(1-x_{AL})^2 (x_{AL} - x_{Ai})}\, d x_{AL}$$

If $\begin{cases} k_x' \to K_x' \\ k_y' \to K_y' \end{cases}$ all $x_{Ai} \Rightarrow x_A^*$, $y_{Ai} \Rightarrow y_A^*$



The previous equations are powerful, but not very easy to solve (you can use numerical integration through!)

We need a few simplified cases

Case 1: Absorption of diluted gas mixture

Dilute ? $\Rightarrow$ usually $x_A < 0.1$ (10 mol%)

What are constant? k_x' k_y', & $1 - x_A \simeq 1 - y_A \simeq 1$

Also $\begin{cases} V = V' \frac{1}{1-y} \simeq V' & \text{constant} \\ L = L' \frac{1}{1-x} \simeq V' & \text{constant} \end{cases}$

If simplify $y = y_{AG}$ $y_i = y_{Ai}$

$$Z = \int_0^z dz = \int_{y_2}^{y_1} \frac{V'}{\frac{k_y' a S}{(1-y_A)_{lm}}} \cdot \frac{1}{(1-y_{AG})^2 (y_{AG} - y_{Ai})} dy_{AG}$$

simplify $\Downarrow$

$$Z = \left[\frac{V}{\frac{k_y' a S}{\cancel{(1-y_A)_{lm}}}} \cdot \frac{1}{\cancel{(1-y)}} \right] \int_{y_2}^{y_1} \frac{dy}{y - y_i}$$

$$= \frac{V}{k_y' a S} \cdot \underbrace{\int_{y_2}^{y_1} \frac{dy}{y - y_i}}_{\ln\left(\frac{y_1 - y_i}{y_2 - y_i}\right)}$$

$$\frac{V}{S} = k_y' a Z \frac{1}{\ln\left(\frac{y_1 - y_i}{y_2 - y_i}\right)}$$

multiply by $(y_1 - y_2)$

$$\frac{V}{S}(y_1 - y_2) = k_y' a Z \boxed{\frac{(y_1 - y_i) - (y_2 - y_i)}{\ln\left(\frac{y_1 - y_i}{y_2 - y_i}\right)}}$$

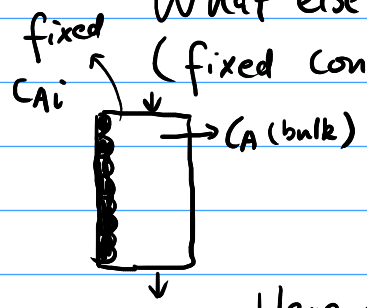
$\Rightarrow$ log mean of $(y_1 - y_i)$ $(y_2 - y_i)$

$$= k_y' a Z \underline{(y - y_i)_m}$$

log mean driving force

What else system has log-mean driving force ? (L21)

(fixed concentration packed bed)



$$Q\,(C_2 - C_1) = k_c' a H\,(C_i - C)_m$$

Here :

$$\frac{V}{S}(y_1 - y_2) = k_y' a Z\,(y_1 - y_i)_m$$

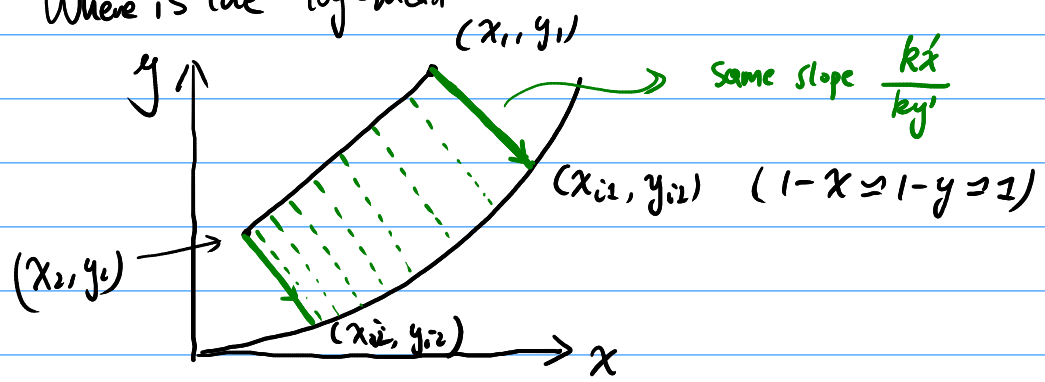
varying y_i interface !

So, for dilute system with const k_y

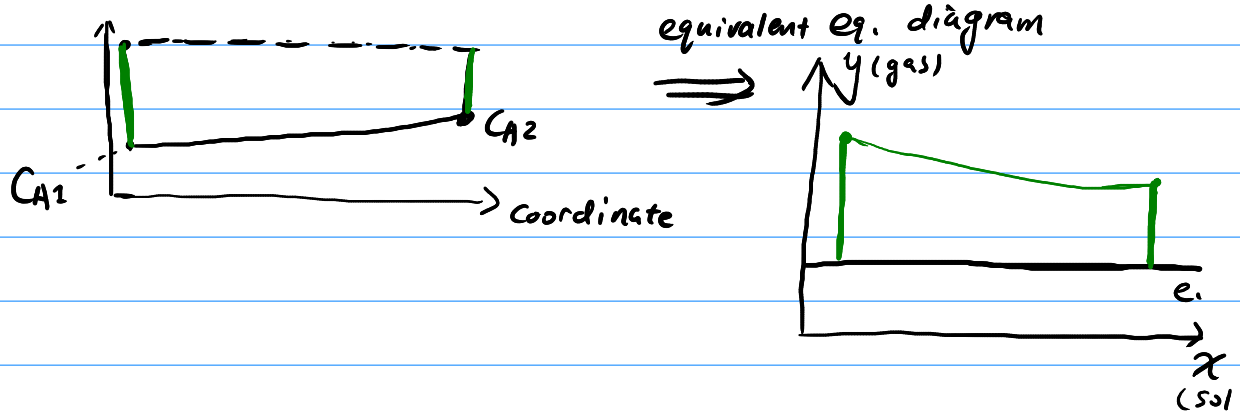
1-phase mass balance

[Flow rate][Conc diff] = [Mass Transfer coeff]·[Aeff]·[log mean D.F.]

Where is the log-mean ?



Compare in fixed conc packed bed



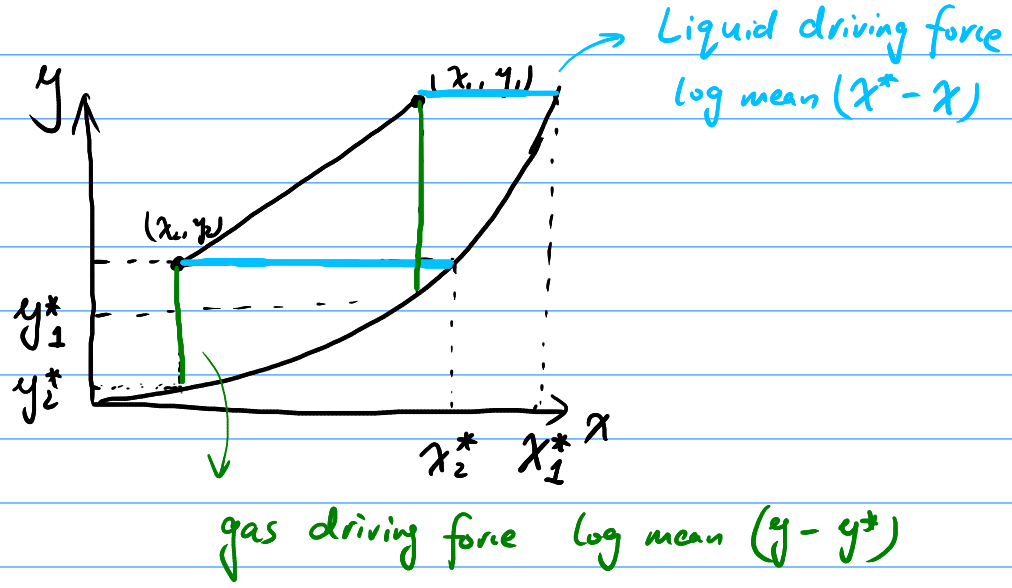
Other forms

Liquid: $\frac{L}{S}(x_1 - x_2) = k_x' a Z (x_i - x)_m$

Using overall coefficients

Gas $\frac{V}{S}(y_1 - y_2) = K_y' a Z (y - y^*)_m$

Liq $\frac{L}{S}(x_1 - x_2) = K_x' a Z (x^* - x)_m$



How do we see V & L?

Usually $\begin{cases} V = \frac{V_1 + V_2}{2} \\ L = \frac{L_1 + L_2}{2} \end{cases}$